INTRODUCTION TO ROBOTICS

WORKSHOP

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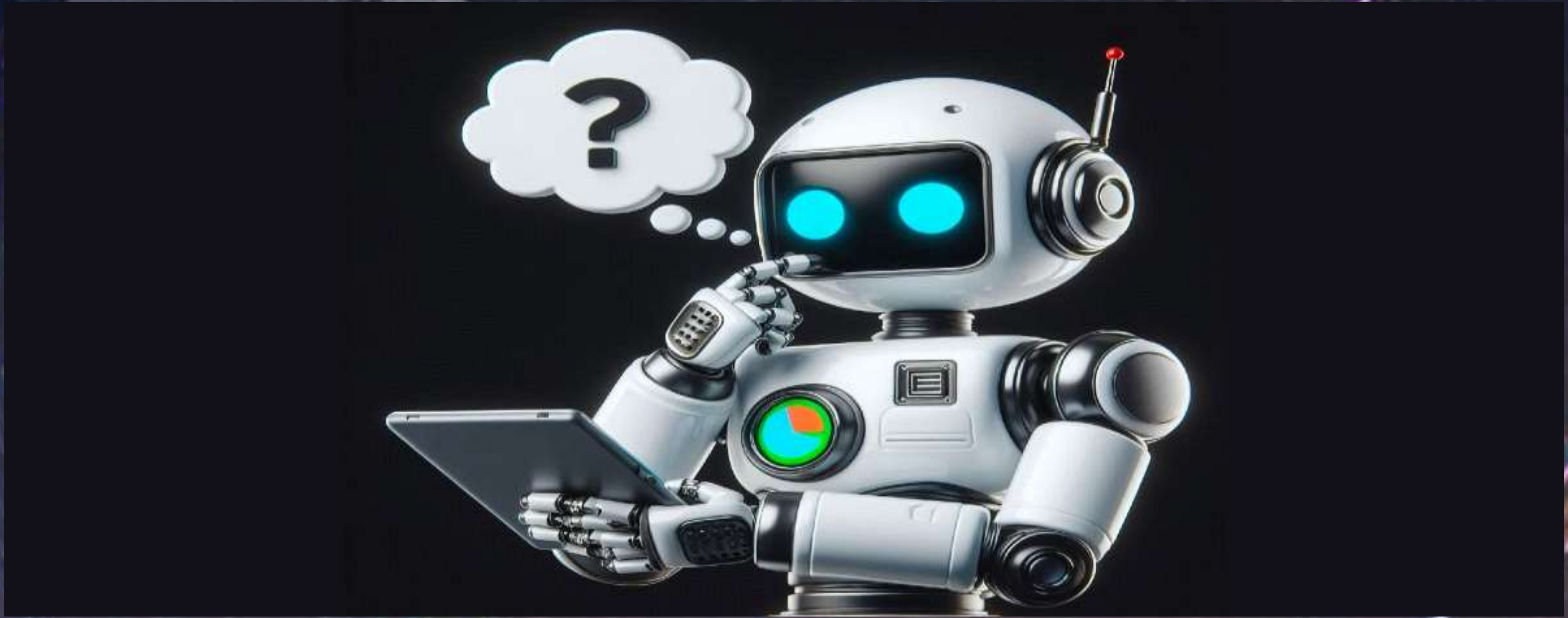
Agenda of the Workshop

Key focusing Areas:

- Introduction to Robotics and IoT technology
- Requirement for Robots technology
- Robotics/ IoT Applications
- Robotics Career Guidance
- Hands-on Project for Robotics (Practical Session)
    - Online tools for IoT/ Robot project
    - Make a simple robot system

What is a Robot...?



What is a IoT …?

- IoT refers to a network of interconnected devices that communicate and exchange data over the internet.
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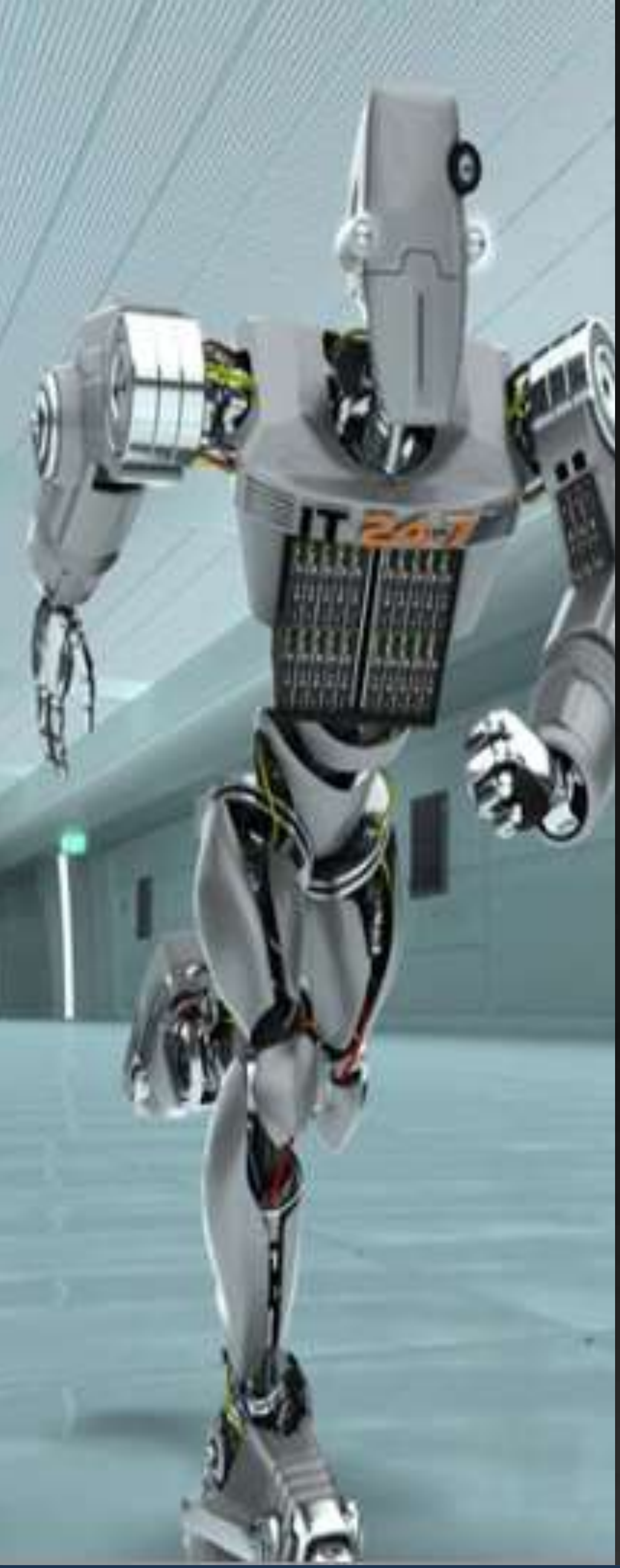


ROBOTICS TECHNOLOGY

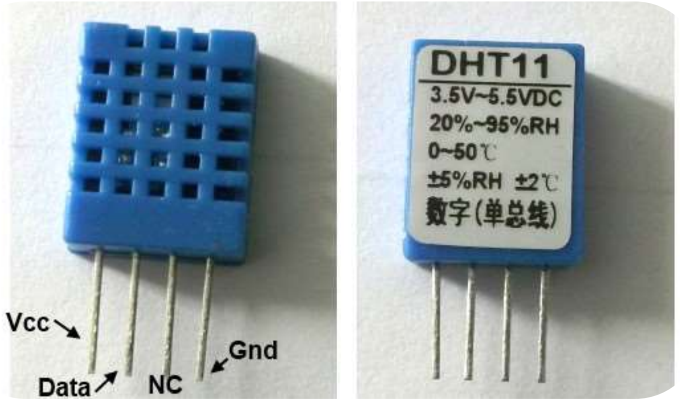
Most industrial robots have at least the following five parts:

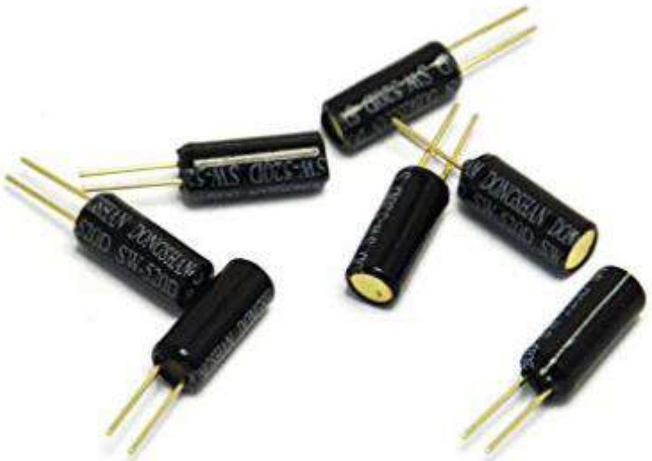
- Sensors
- Actuators
- Controllers
- Arms

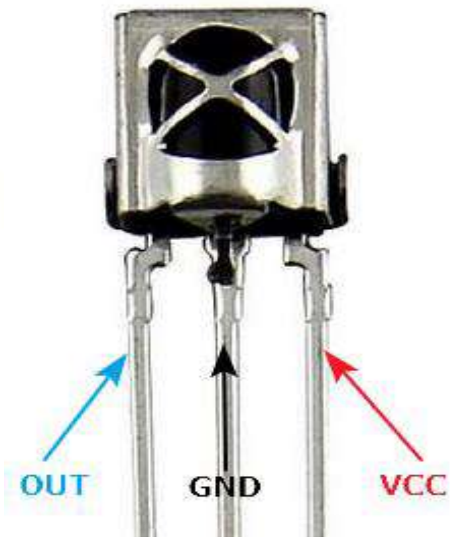
Sensors

Used for collecting data from the physical environment.

- PIR Sensors
- IR Sensors
- Ultrasonic Sensors
- Accelerometers
- Magnet Sensors
- Gas Sensors
- Temp Sensors
- Humidity Sensors



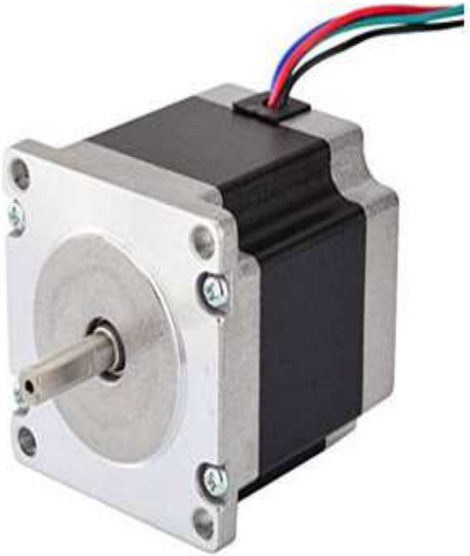




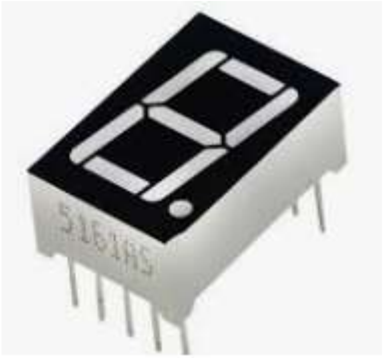
Actuators

These are used to perform actions based on received instructions.

- LED
- LCD Display
- Motors
- Buzzer
- Relay
- Valve






Controllers

Controllers (such as microcontrollers or processors) are a critical component of Robotics/ IoT devices, as they function as the "brain" of the system.

- PIC microcontroller
- Atmel microcontroller
- Intel microcontroller
- EPSON microcontroller
- PLC
- Raspberry Pi

Robot Arms/ Hardware

Robot Arms are mechanical devices designed to simulate the movements and functionalities of a human arm.

Robotics Applications

- Assembly lines robots
- Welding robots
- Painting robots
- Healthcare Surgical Robots
- Robotic vacuum cleaners

IoT Applications

- Smart Homes
- Smart City
- Self-driven Cars
- Wearable health trackers
- IoT Retail Shops
- Agriculture
- Safety and Security

Current Trends and Opportunities in Sri Lanka

- **Automation in Manufacturing:**

  Companies in Sri Lanka are accepting robotic automation to improve efficiency in production lines.

- **Agricultural Robotics:**

  Development of robotics solutions for care farming, such as automated harvesting and crop monitoring.

- **Healthcare Robotics:**

  Emerging interest in robotic-assisted surgery and rehabilitation devices in the healthcare sector.

- **Robotics Research and Education:**

  Universities and institutions are focusing on robotics education and projects, offering more courses and competitions to nurture talent.

Current Trends and Opportunities in Japan

- **Industrial Robotics:(example)**

  Japan is a global leader in robotics, with companies like **FANUC, Yaskawa, and Kawasaki producing advanced industrial robots**.

- **Service Robots: (example)**

  Innovations in hospitality, retail, and elder care robots, such as Pepper and Paro.

- **Healthcare Robots: (example)**

  Extensive use of robots for elder care, surgical assistance, and rehabilitation.

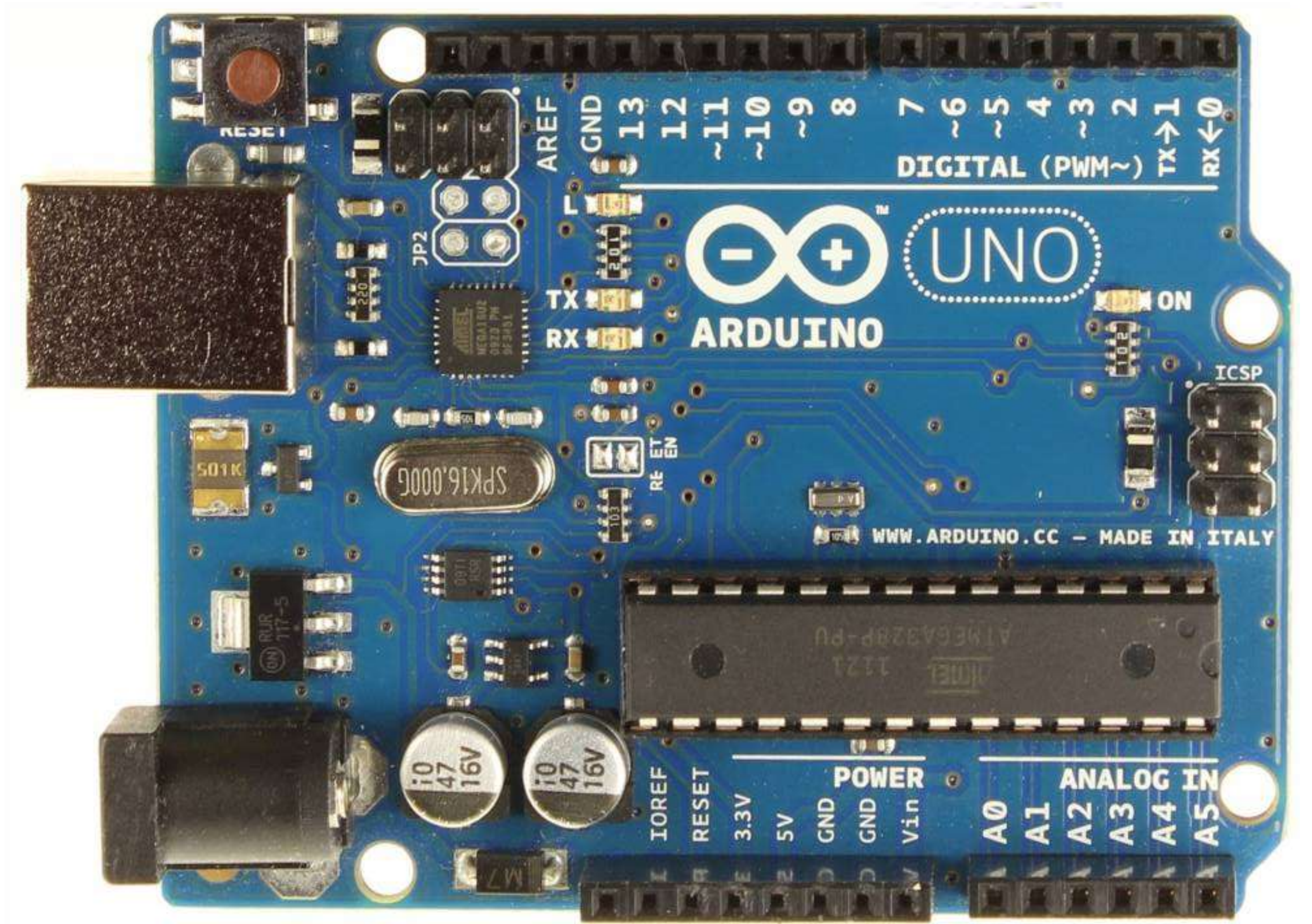
- **AI Integration:**

  Robotics combined with artificial intelligence for autonomous decision-making in various industries.

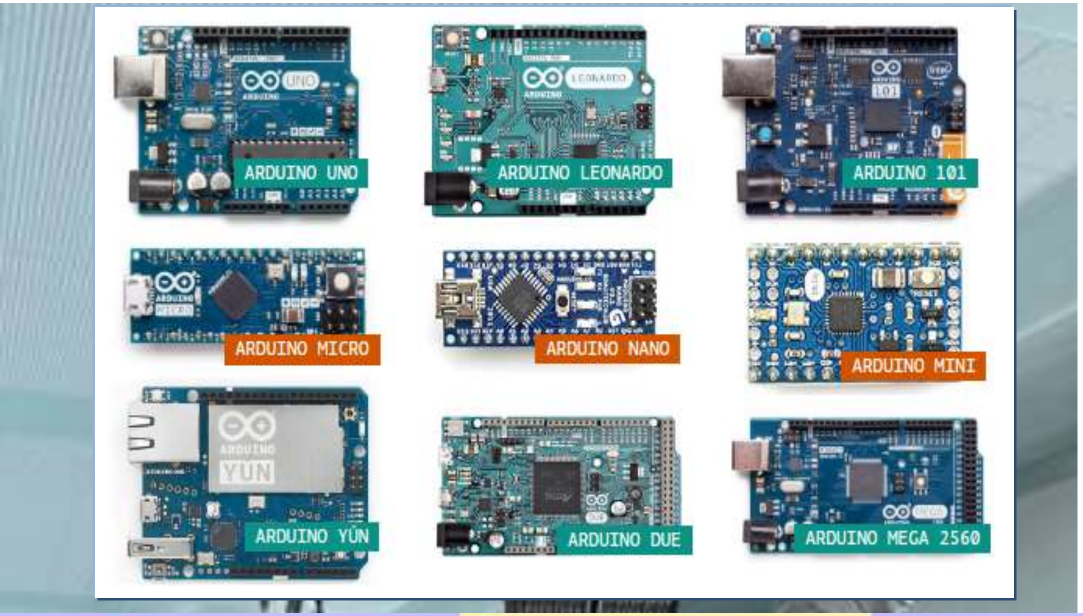
Microcontroller Development Boards

- A microcontroller development board is a circuit board that can used to program and test with a specific microcontroller boards capabilities

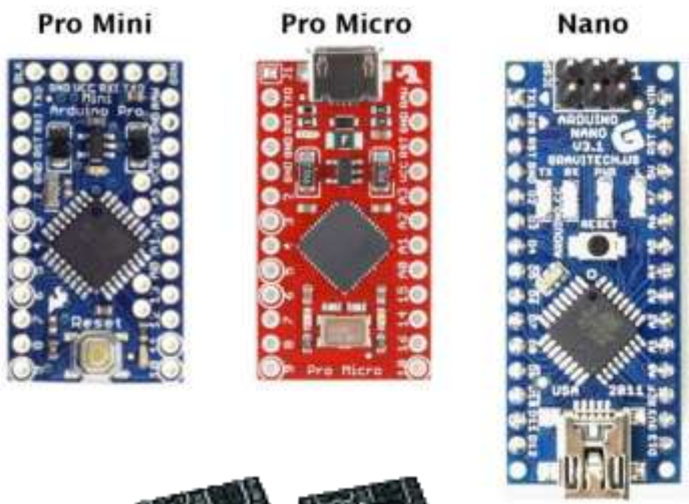
- Raspberry Pi Board
- ESP8266 Board
- PIC Board
- PLC Board
- **Arduino Board**

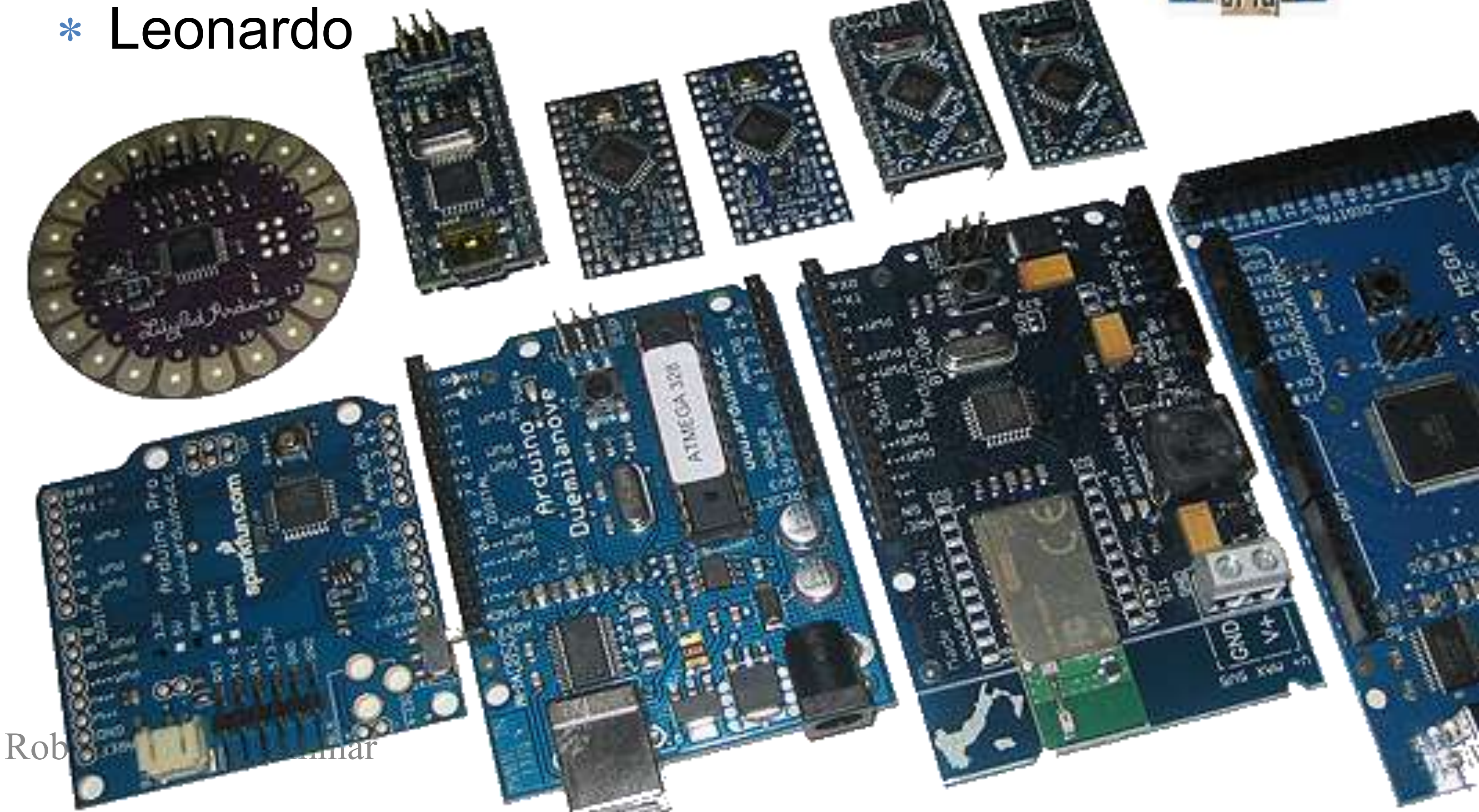


Arduino Boards



- Due
- Mega
- Uno
- Nano
- Mini
- Micro
- Leonardo

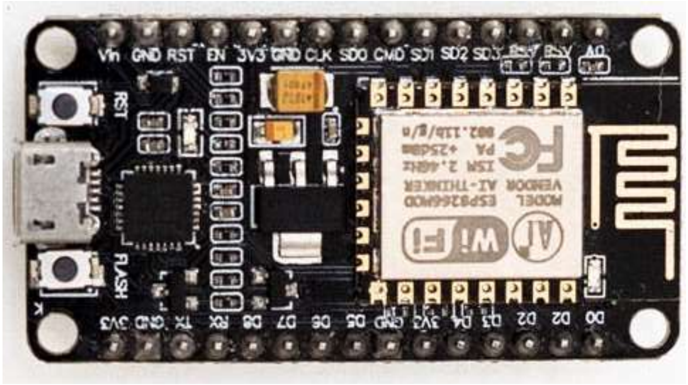


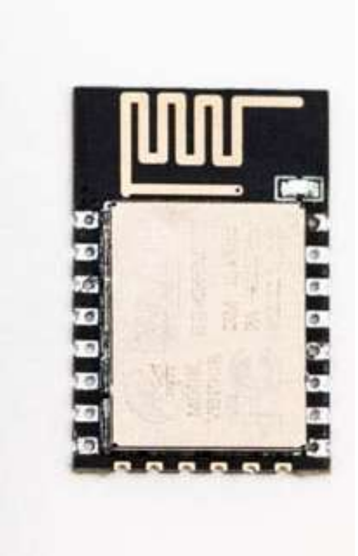


Select suitable Boards for Robotics & IoT

- Wemos D1
- NodeMCU
- ESP32 Boards

- Nano 33 IoT
- MKR WiFi 1010





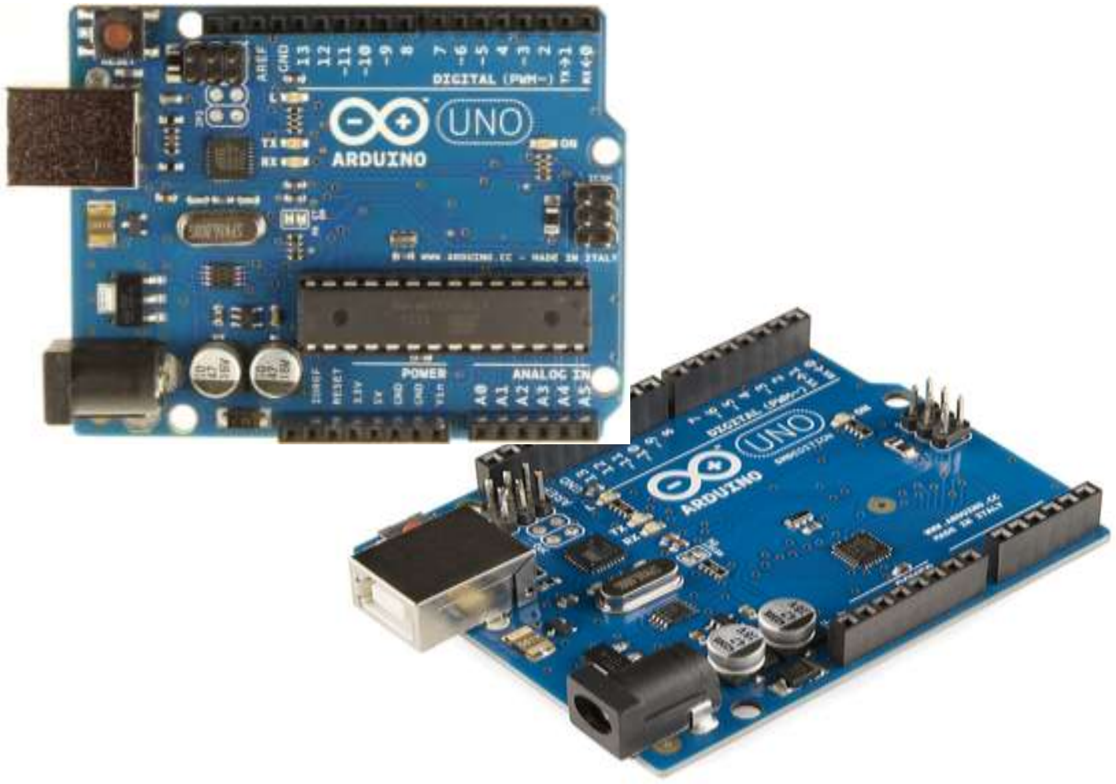
Arduino Uno Board Vs Mega Board

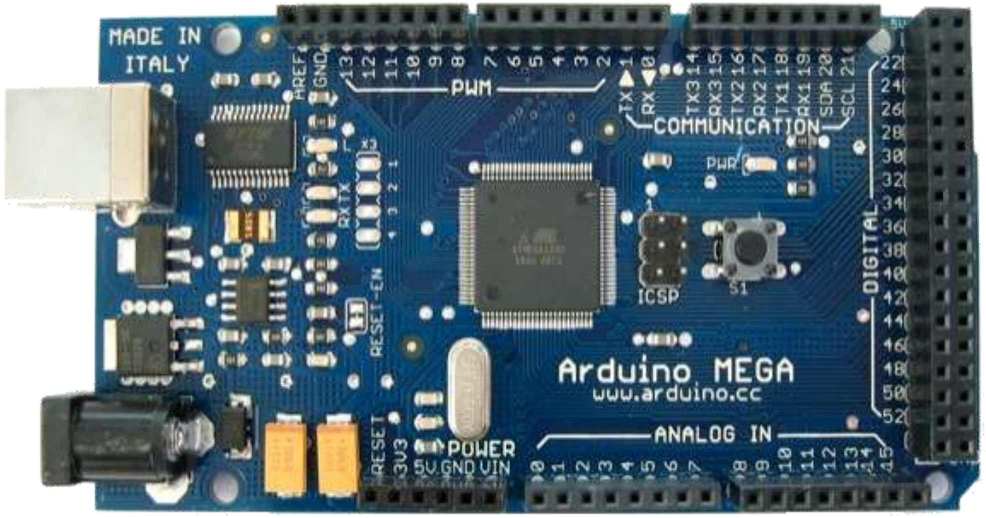
UNO – **Atmega 328P chip**

- Digital pins -14
- Memory - 32 kb
- Ram- 2 kb

Mega – **Atmega 2560 chip**

- Digital pins -54
- Memory -256 kb
- Ram- 8kb





Write control Programme - Cording Vs Scratch

Arduino

```
void setup() {
  pinMode(13, OUTPUT);
}

void loop() {
  digitalWrite(13, HIGH);
  delay(1000);
  digitalWrite(13, LOW);
  delay(1000);
}
```

S4A

Write control Programme – Scratch S4A

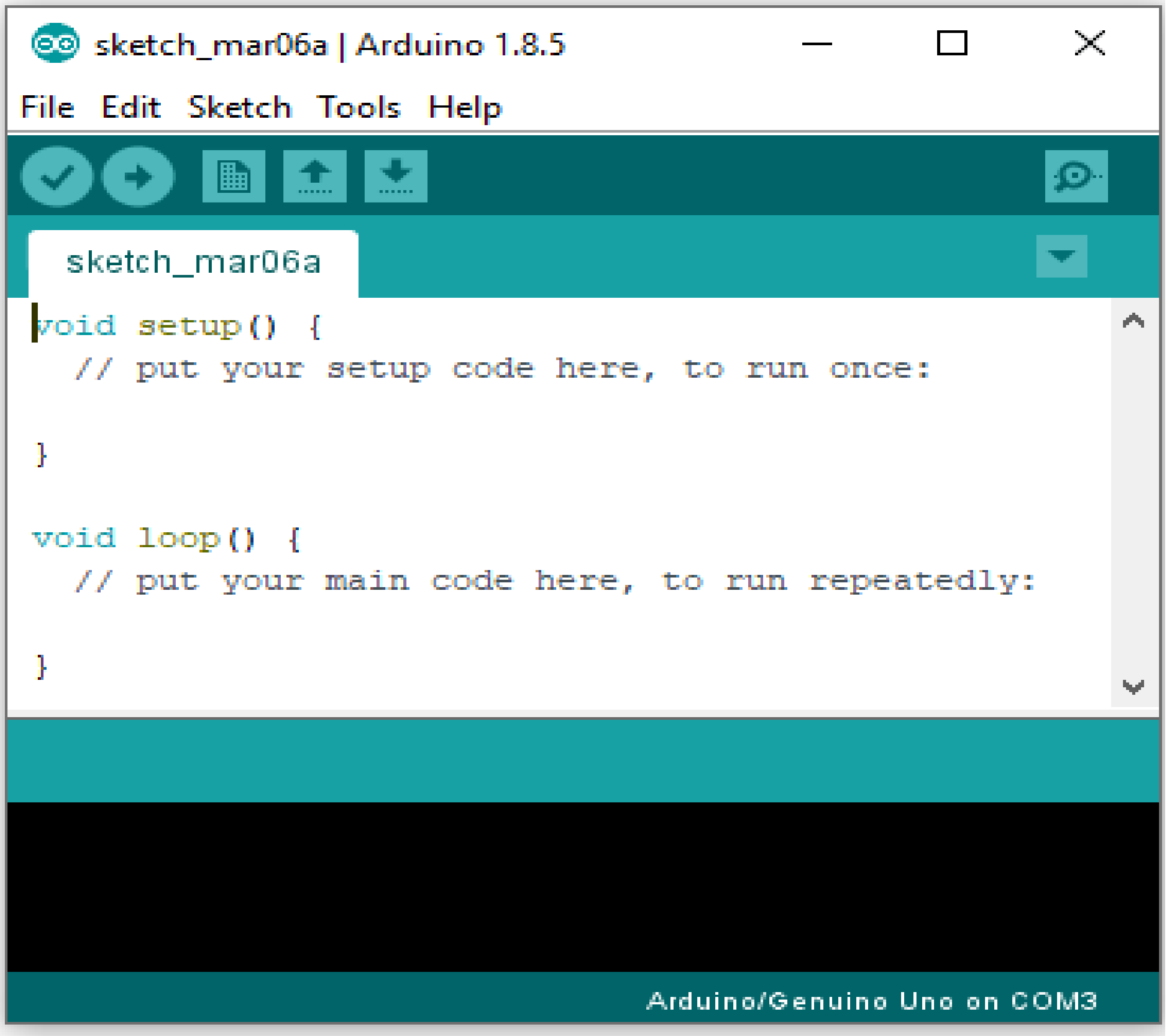
- Code is very similar to Arduino
- Block-based, so less likely to make mistakes

S4A

Scratch for Arduino

Programming Sketches - Cording

```
void setup() {
        pinMode(13, OUTPUT);
        }

void loop() {
    digitalWrite(13, HIGH);
    delay(1000);
    digitalWrite(13, LOW);
    delay(1000);
  }
```



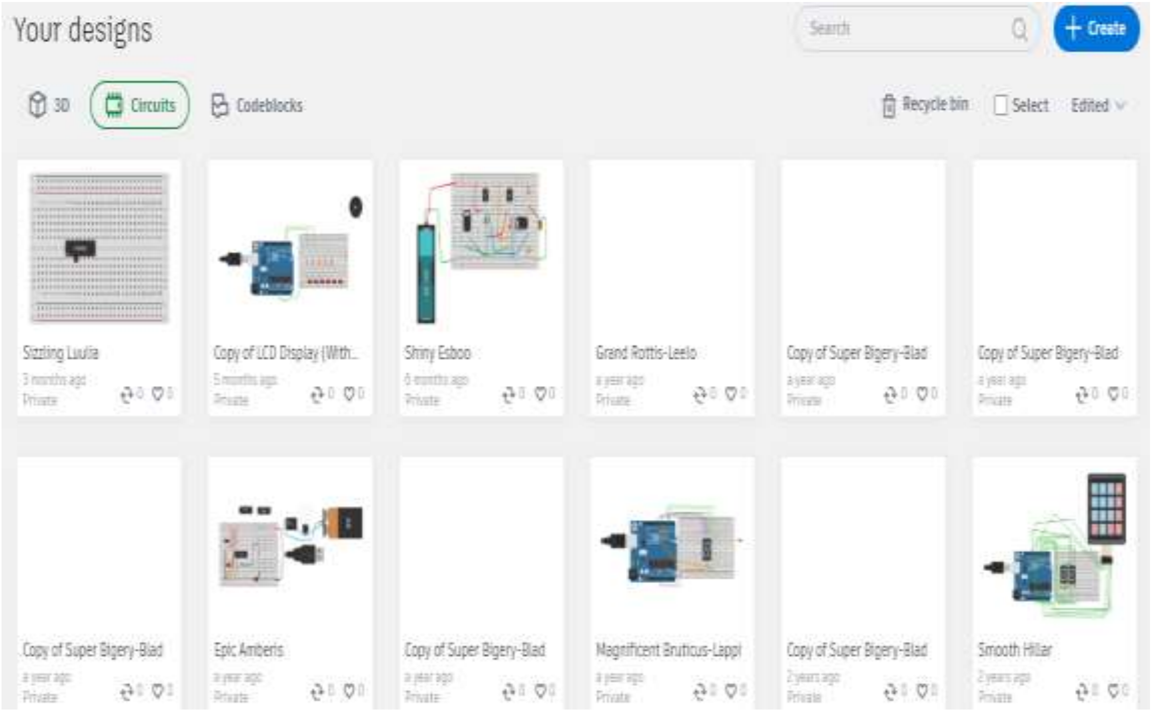
Online Tools:

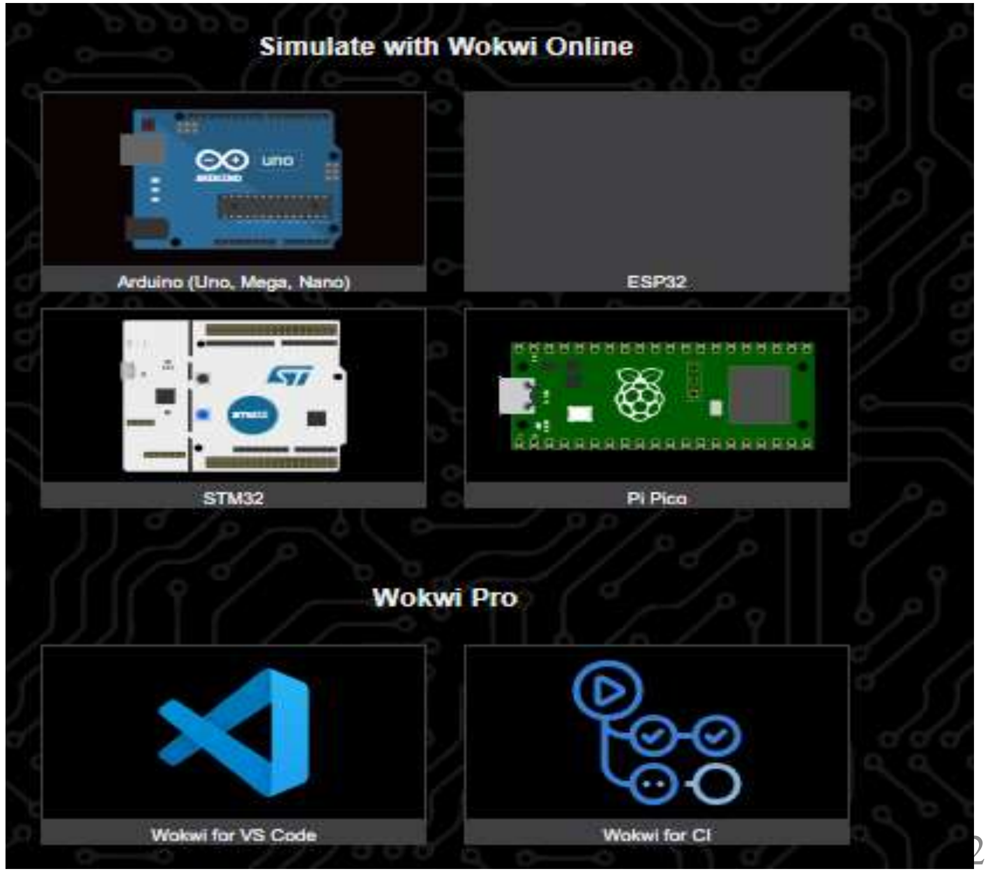
https://tinkercad.com

https://wokwi.com/

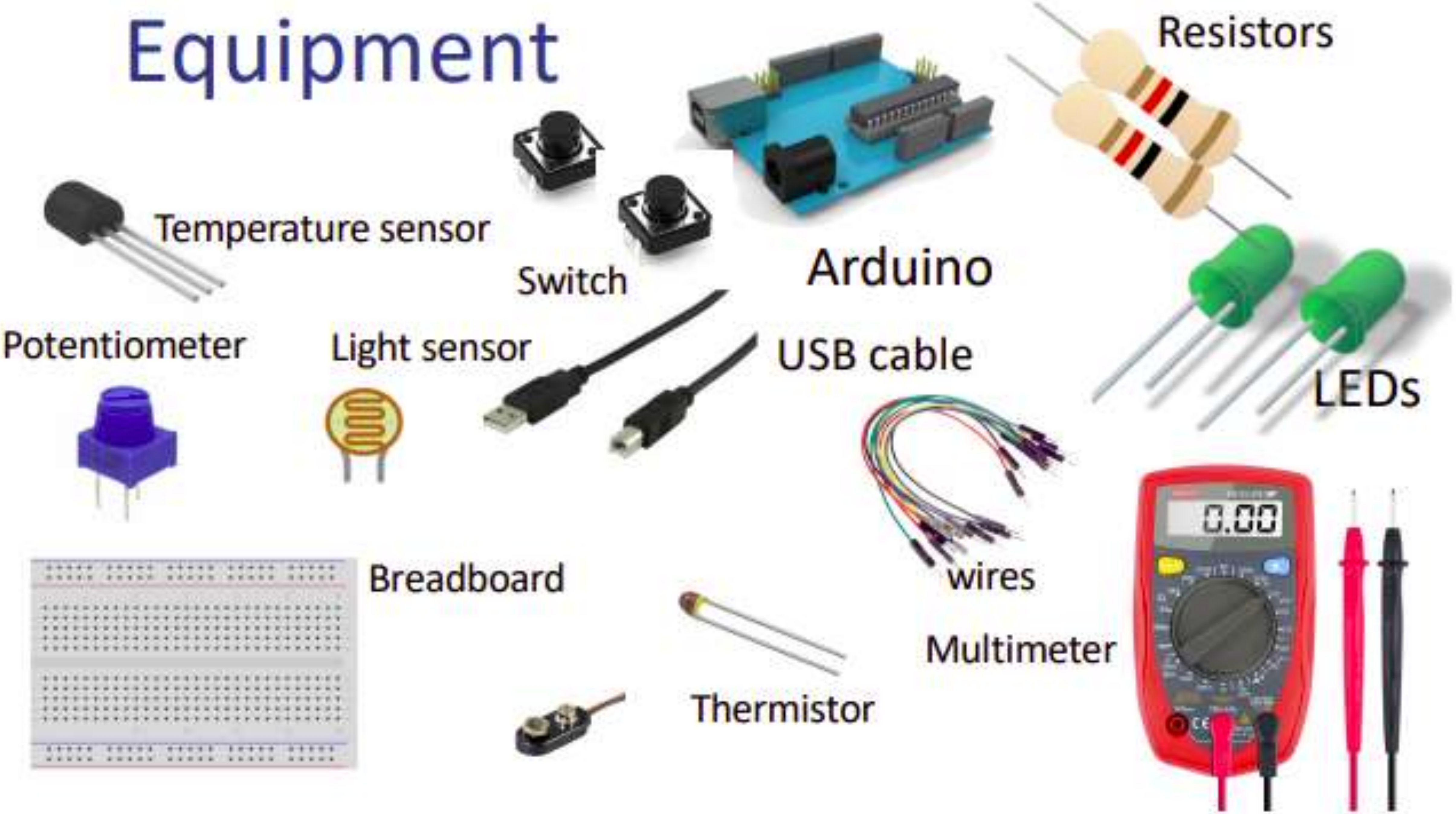
AUTODESK Tinkercad

WOKWI
Simulate IoT Projects in Your Browser
Discord Community LinkedIn Group



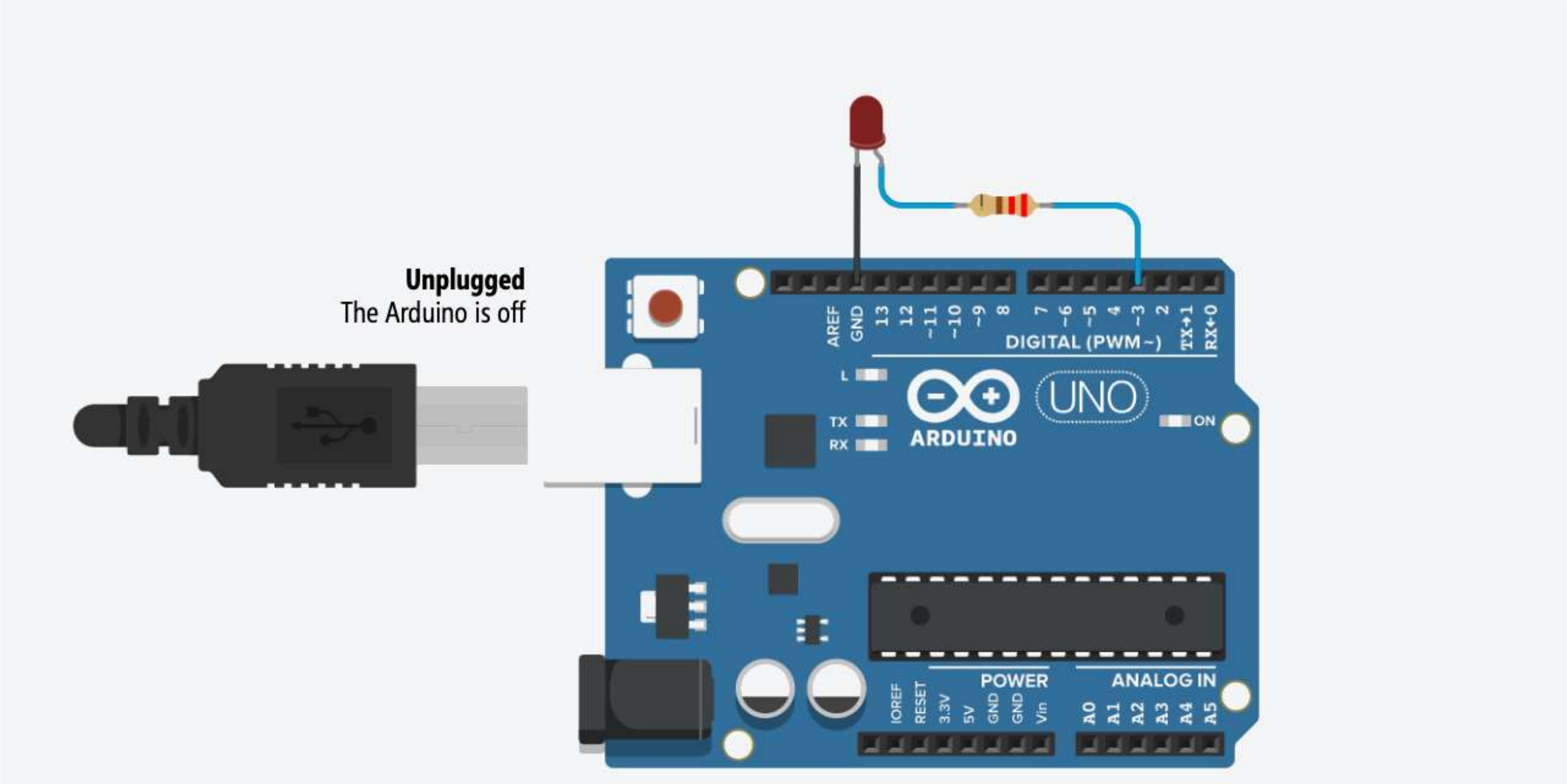


Equipment


Resistors
Temperature sensor
Switch
Arduino
Potentiometer
Light sensor
USB cable
LEDs
Breadboard
wires
Multimeter
0.00
Thermistor

Hands-on Project for Robotics (Practical Session)

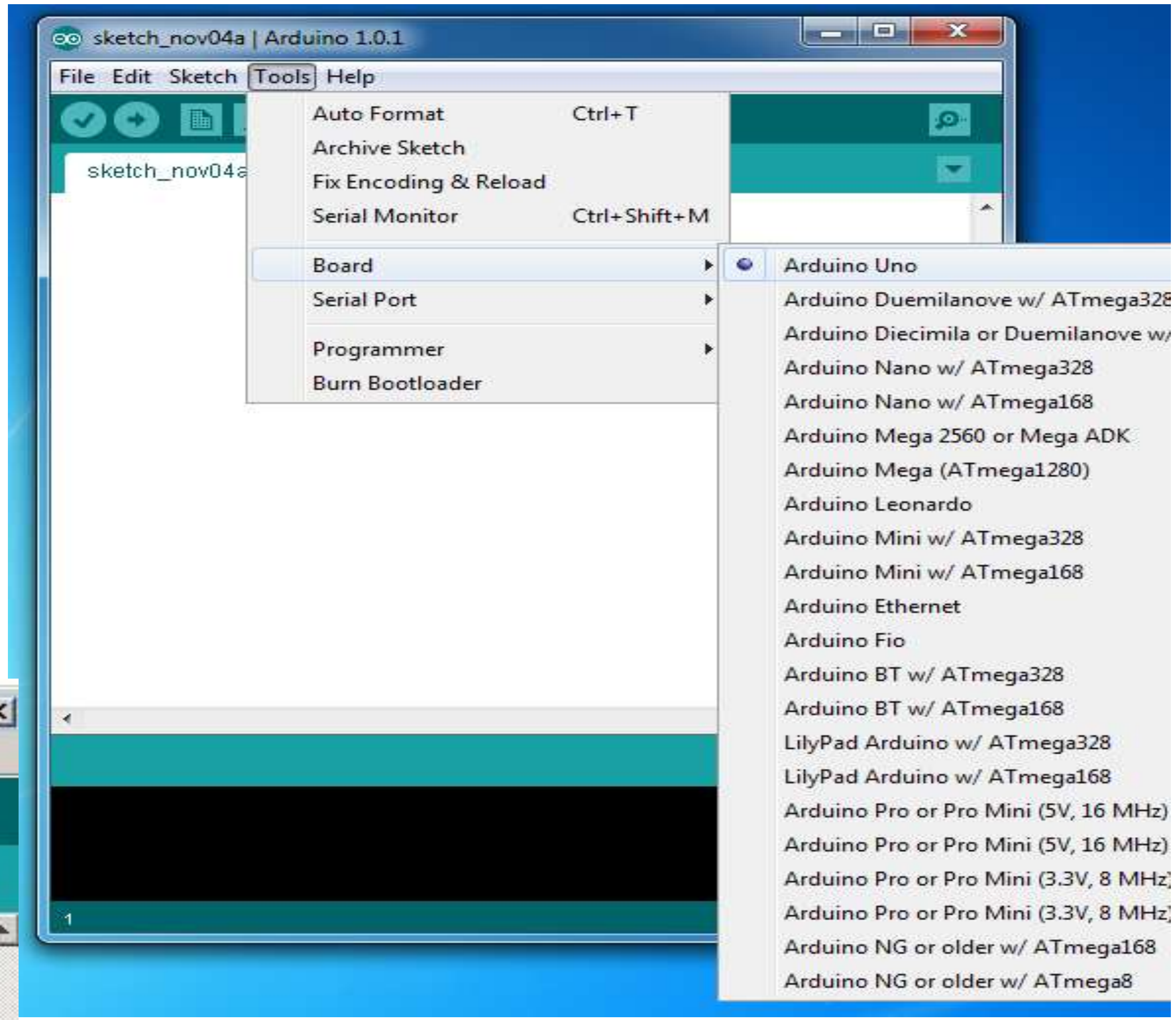
LED Blinking circuit

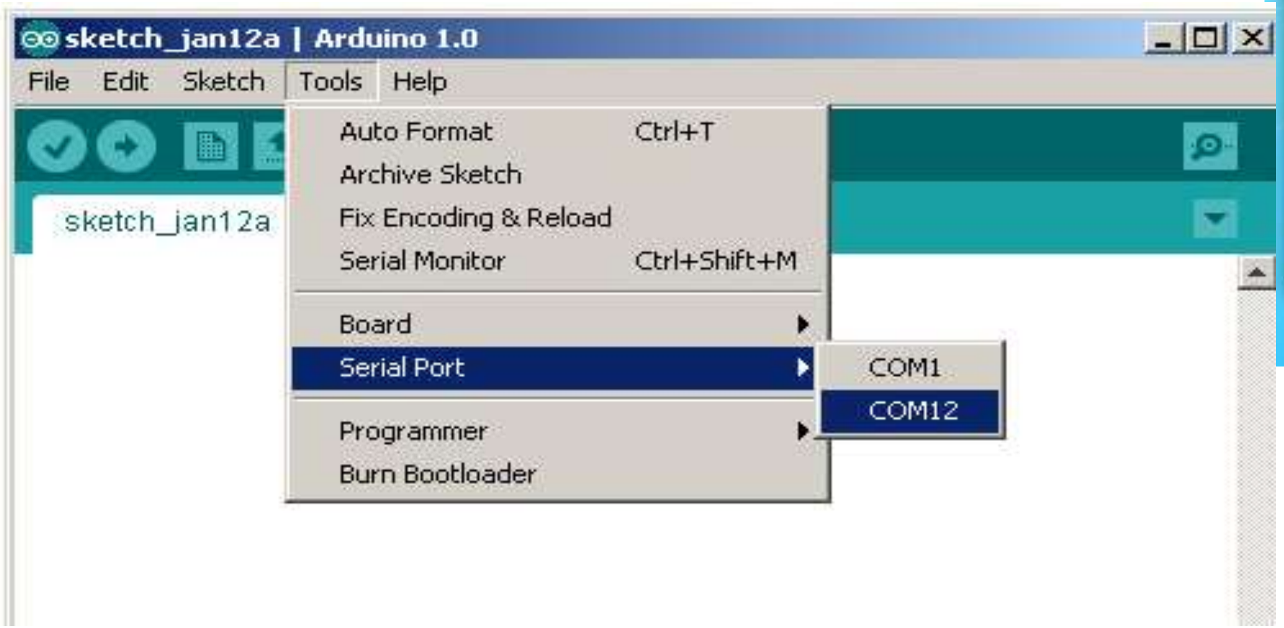


Select Board and Port

- Select board
  Tools→ Board→ Select board

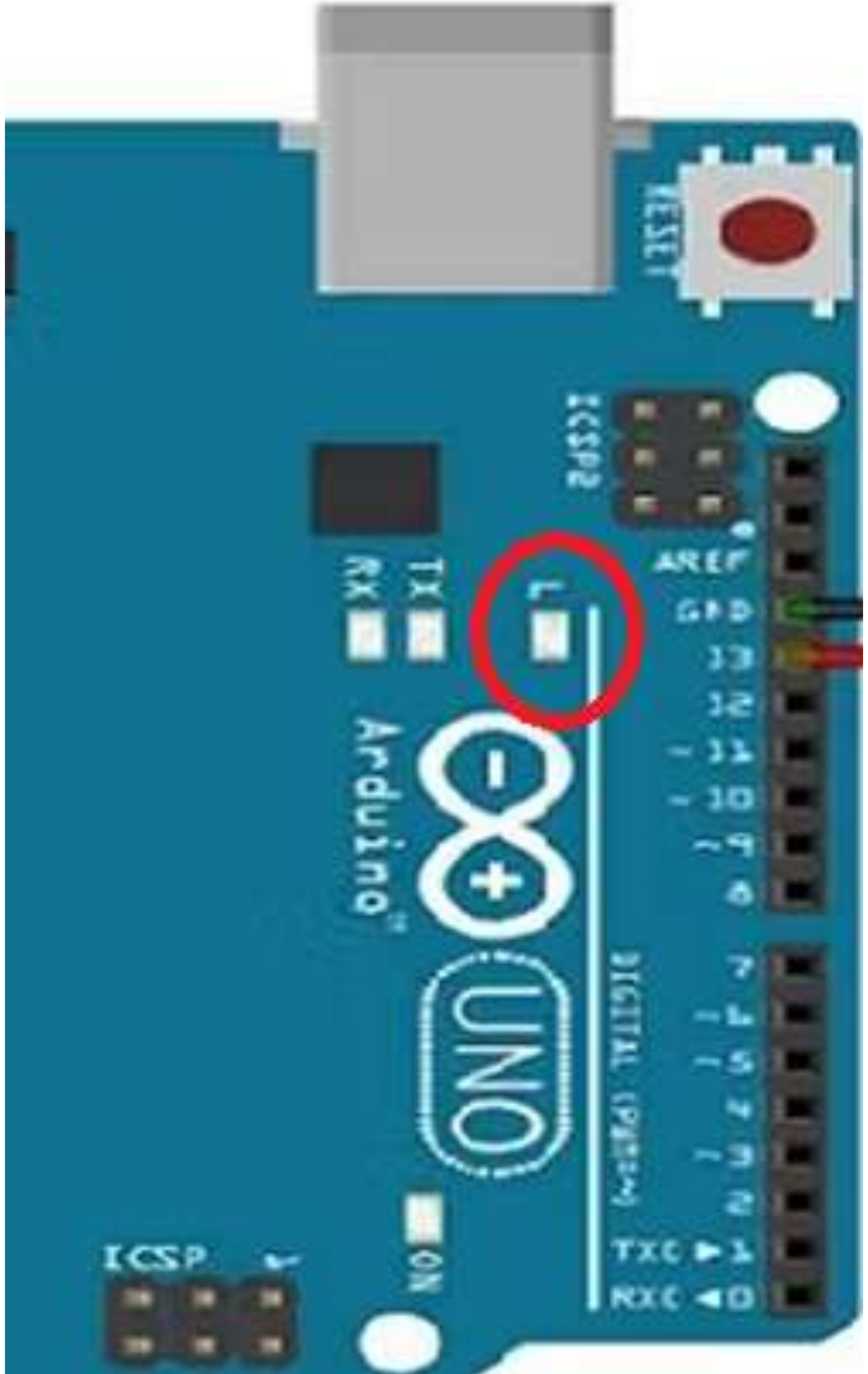
- Select port
  Tools → Serial Port→ COM xx





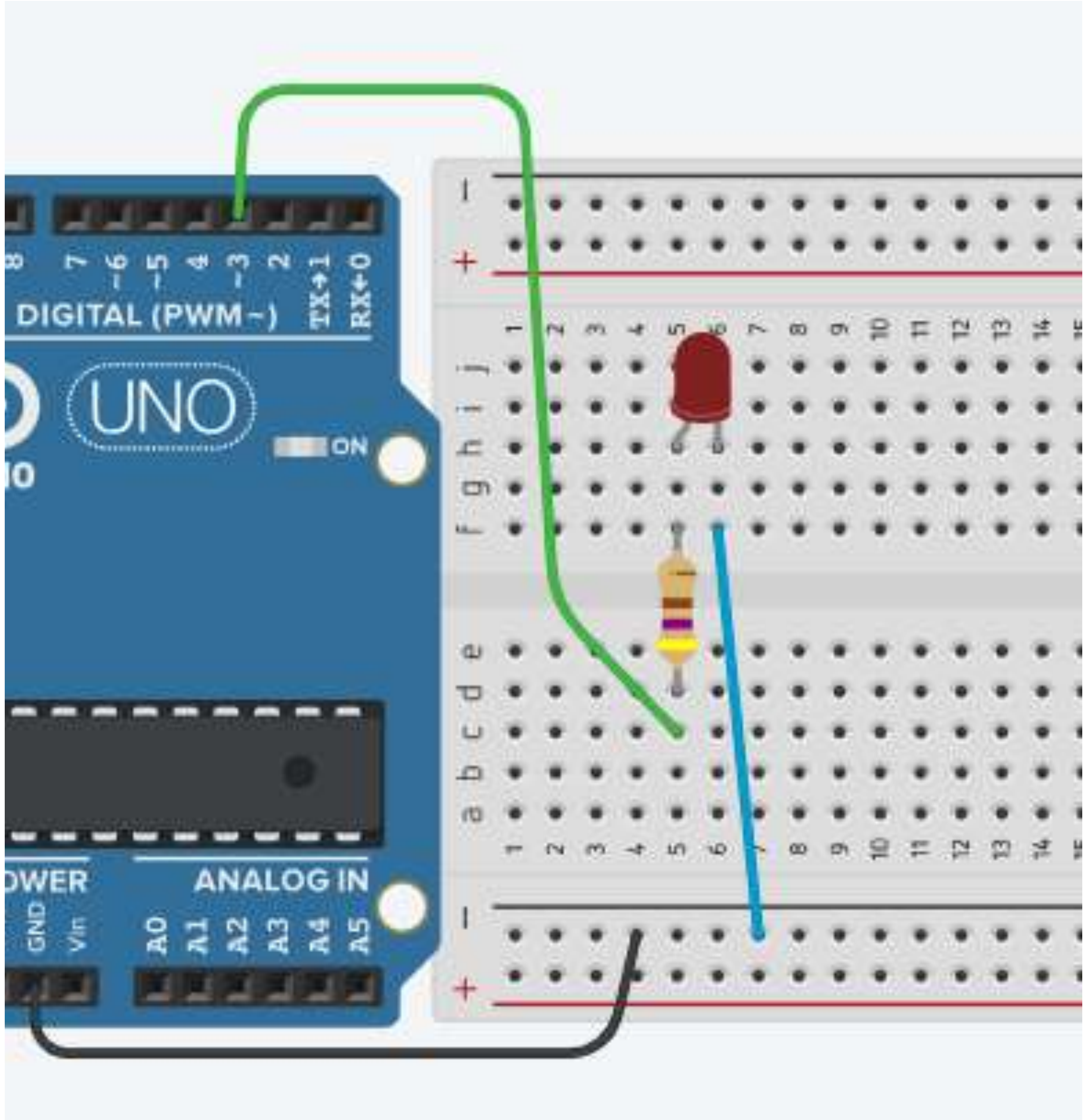
Board Testing -01

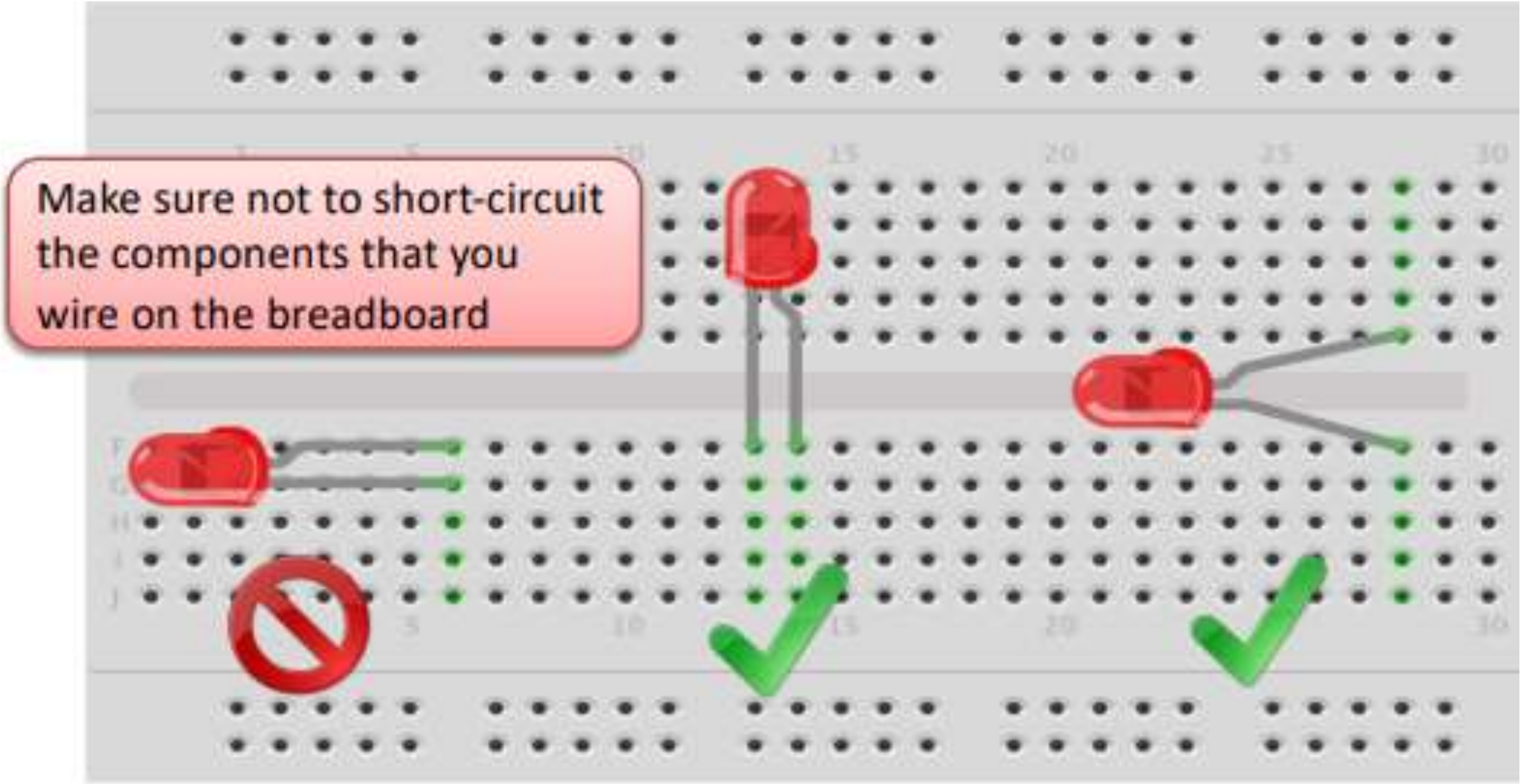
```
void setup() {
        pinMode(13, OUTPUT);
        }

void loop() {
   digitalWrite(13, HIGH);
   delay(1000);
   digitalWrite(13, LOW);
   delay(1000);
  }
```



LED blinking circuit on Bread Board

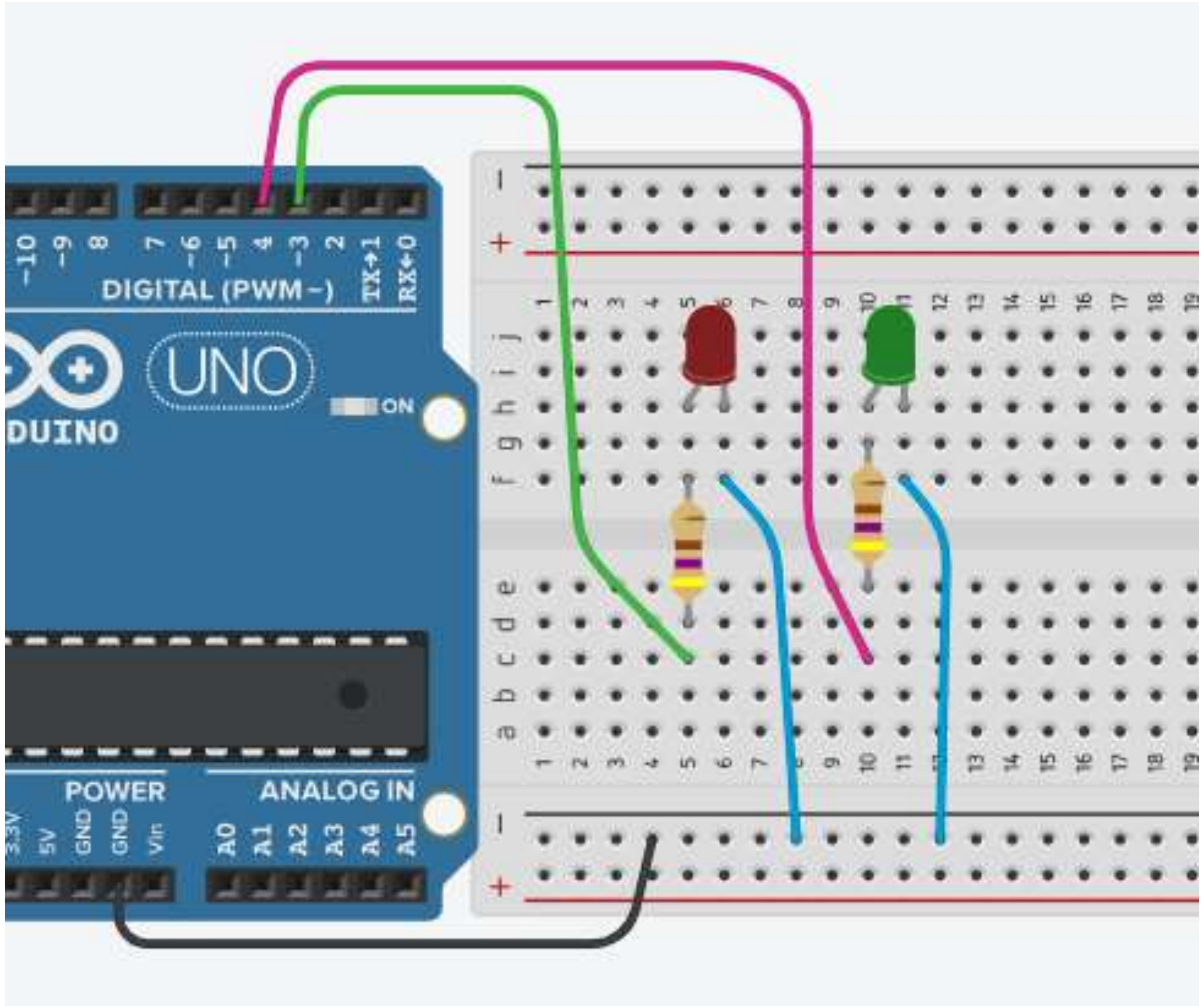
```
void setup() {
        pinMode(3, OUTPUT);
        }

void loop() {
    digitalWrite(3, HIGH);
    delay(1000);

    digitalWrite(3, LOW);
    delay(1000);
   }
```




Make sure not to short-circuit the components that you wire on the breadboard

2 LED blinking circuit

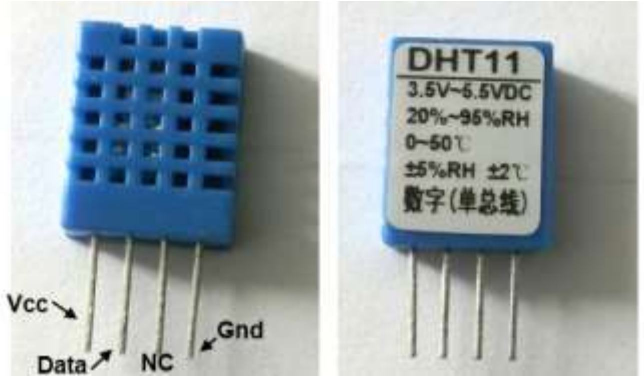
```
void setup() {
     pinMode(3, OUTPUT);
     pinMode(4, OUTPUT);
               }

void loop() {
      digitalWrite(3, HIGH
      digitalWrite(4, LOW);
      delay(1000);

      digitalWrite(3, LOW);
     digitalWrite(4, HIGH);
      delay(1000);

     }
```



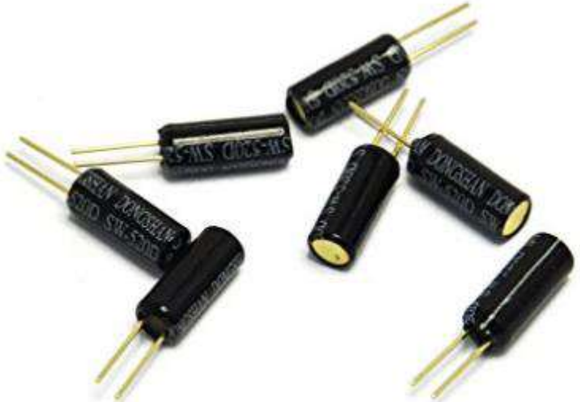
sensors and actuators



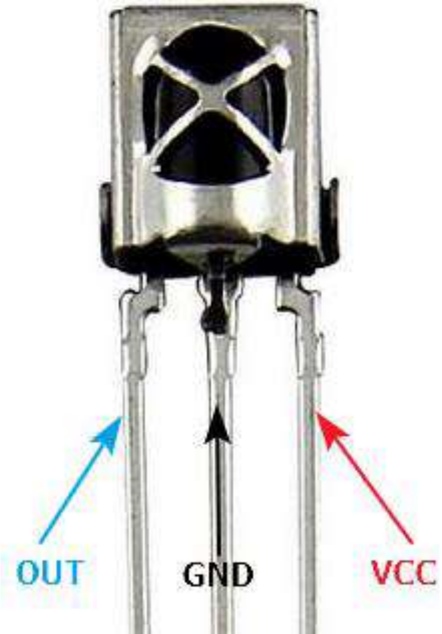


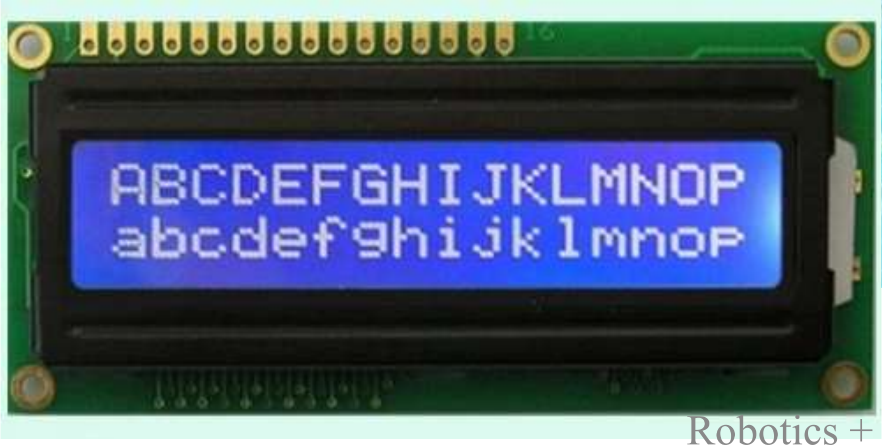


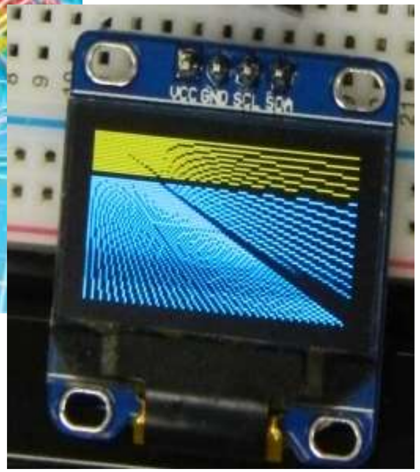




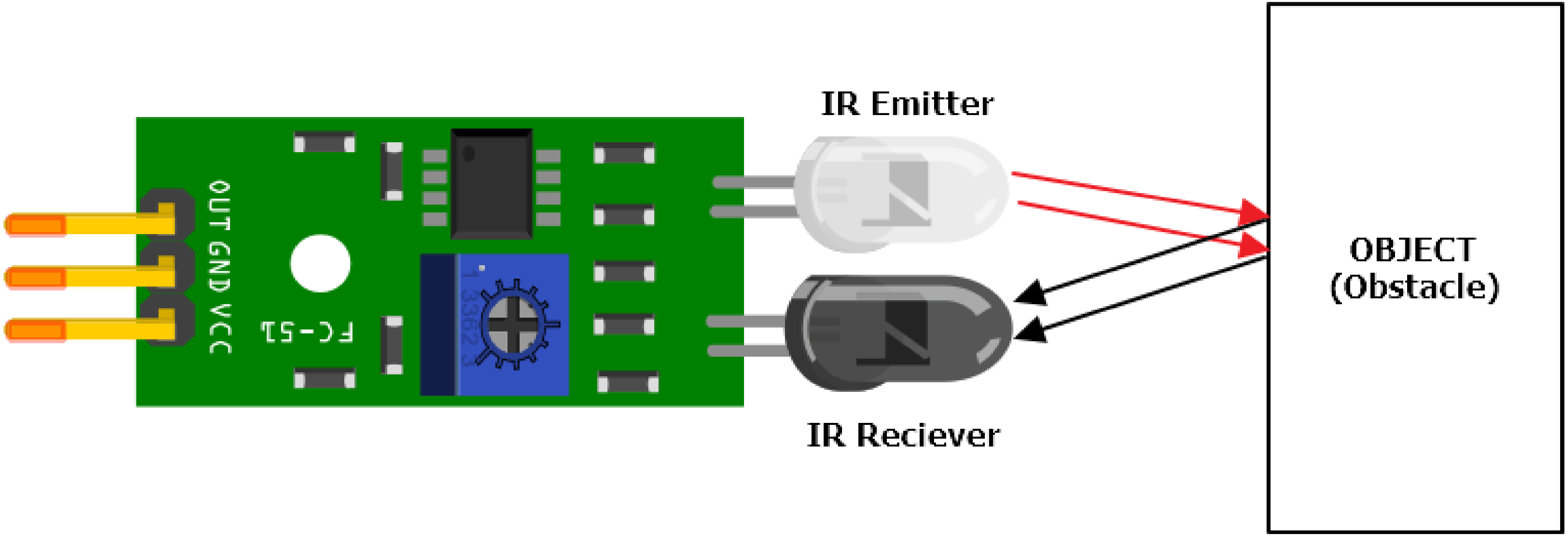






IR Proximity Sensor - Digital

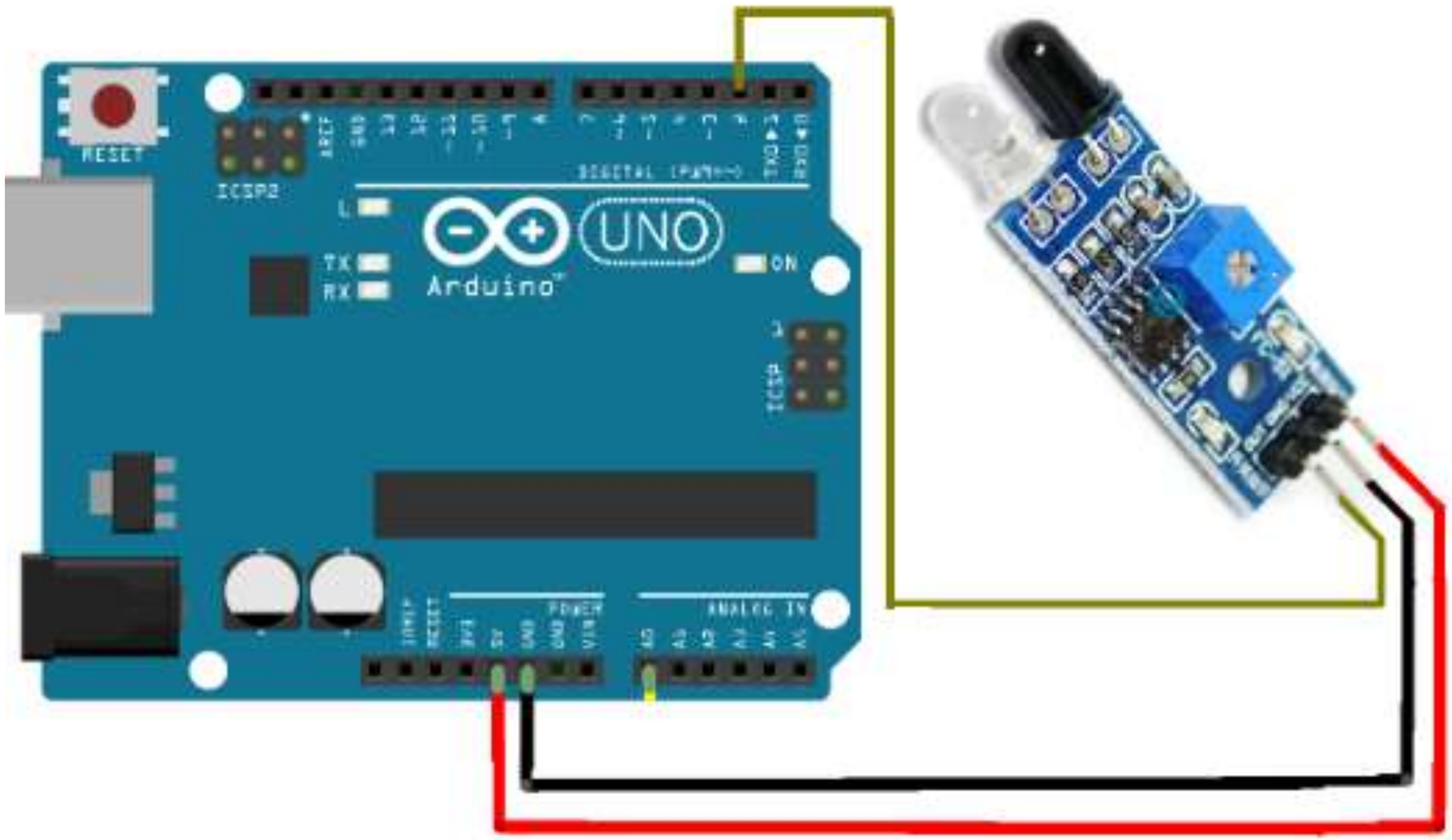


VCC (+)– Power Supply (+5V) pin
GND (-)– Ground pin
OUT (S)– Output pin

IR Proximity Sensor configuration

VCC – Power Supply (+5V) pin
GND – Ground pin
OUT – Output pin – **pin 2**

- Keep your LED circuit as it is
- Connect IR Proximity Sensor to **pin 2**



IR Proximity Sensor configuration programme

```
int Proxpin = 2;
int led1 = 3;
int sensorvalue =0;

void setup() {
  pinMode(led1, OUTPUT);
  pinMode(Proxpin, INPUT);
  }
```

```
void loop() {
sensorvalue = digitalRead(Proxpin);

if(sensorvalue == LOW) {
    digitalWrite(led1, HIGH);
  }
  else{
    digitalWrite(led1, LOW);
  }
 }
```

Activity 01:

IR Proximity Sensor + 2 LEDs Control System

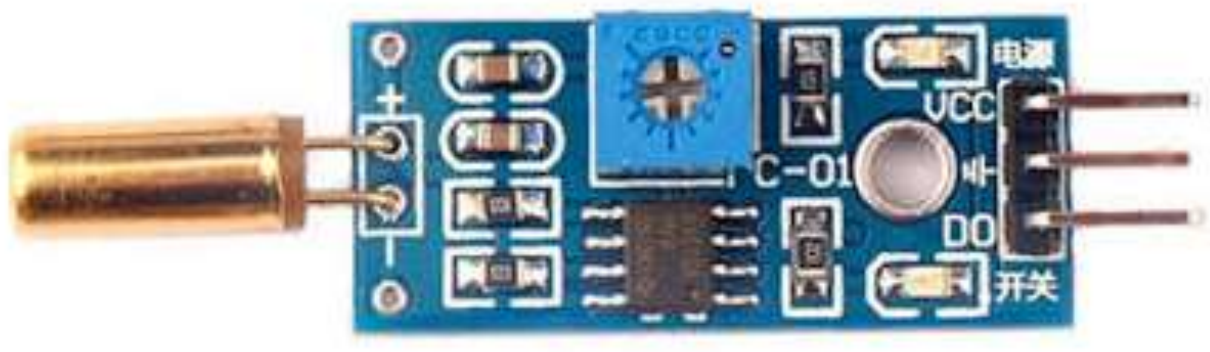
🎯 Objective

Control both LEDs using IR proximity sensor:

- LED 1 → turns ON when object is detected
- LED 2 → turns ON when no object is detected

Tilt Sensor - Digital

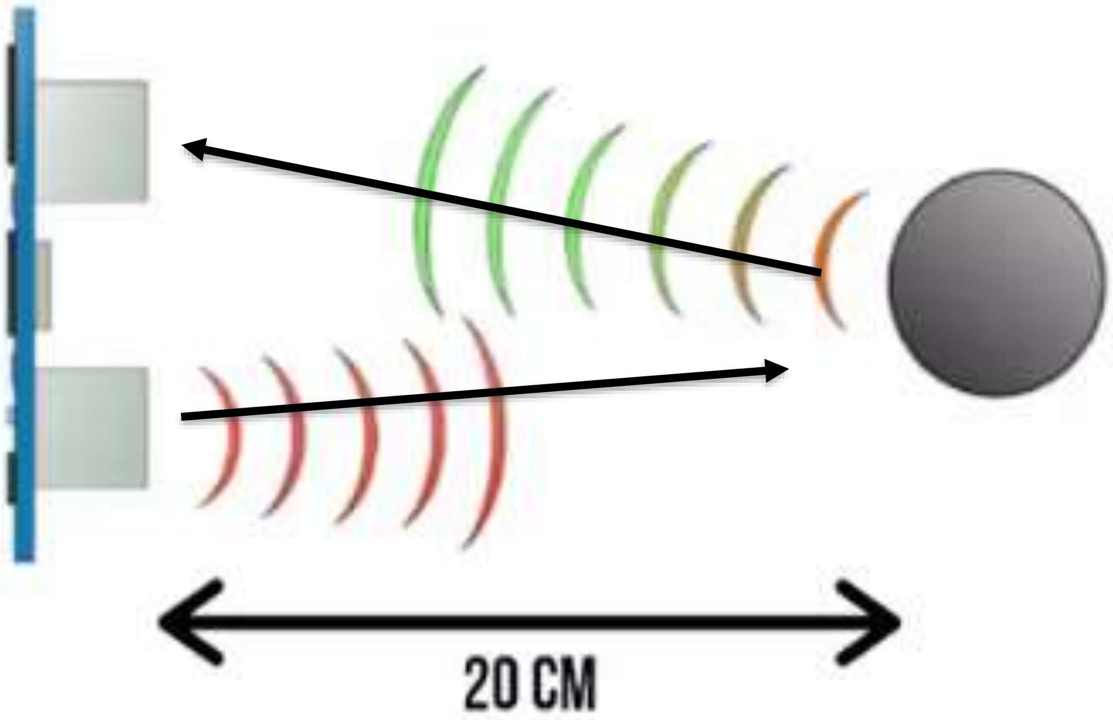
A device that measures the **Vibration,** slope, angle of an object relative to gravity



VCC (+)– Power Supply (+5V) pin
GND (-)– Ground pin
OUT (S)– Output pin

Ultrasonic Sensor Module

VCC – Power Supply (+5V) pin
GND – Ground pin
Echo – Input pin
Trig - Output pin



SPEED OF SOUND:

$v = 340$ m/s

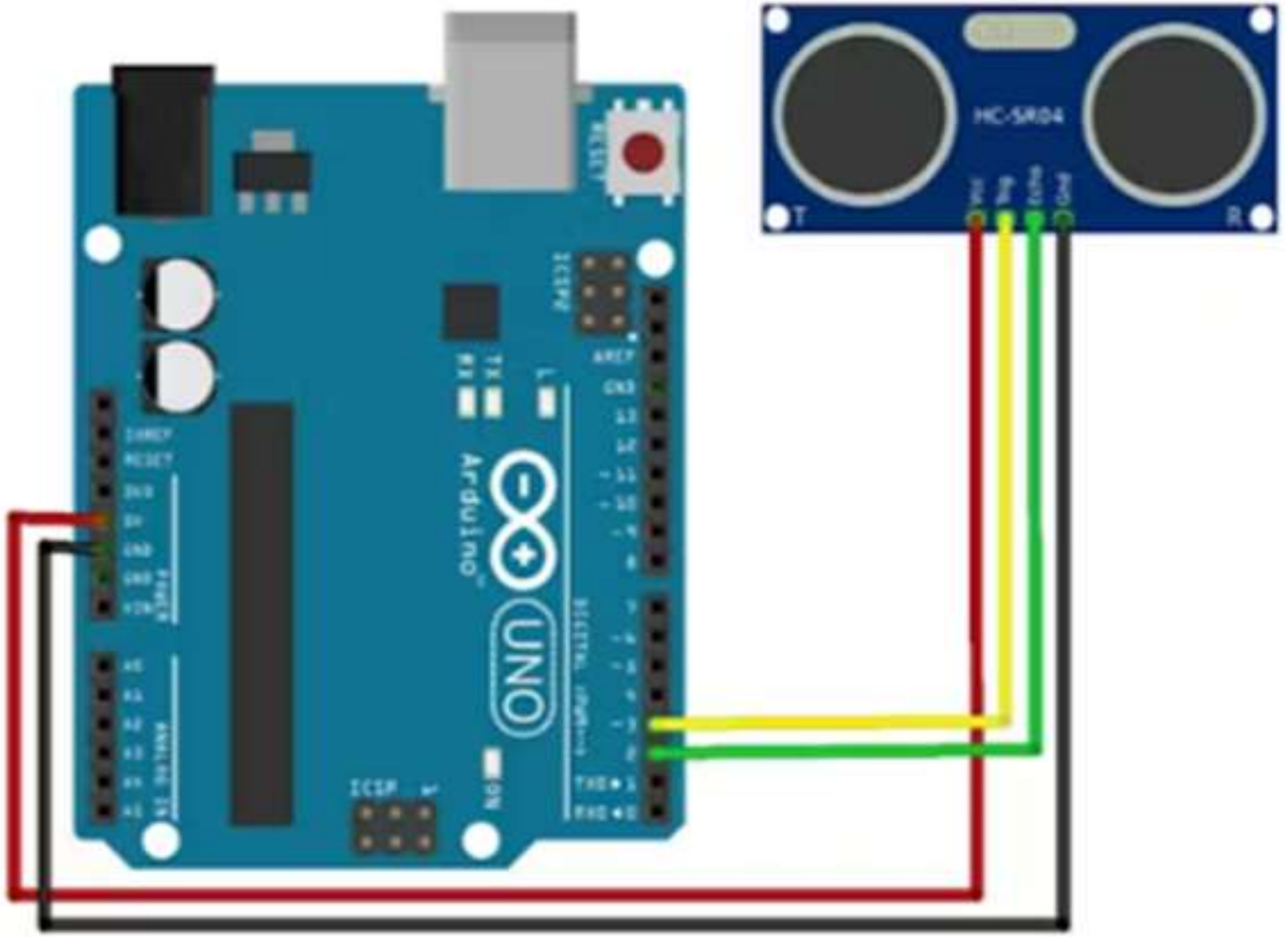
TIME = DISTANCE/SPEED

$t = s/v = 20/0.034$

$= 588$ us

$s = t \times 0.034/2$

Distance measuring programme



VCC – Power Supply (+5V) pin
GND – Ground pin
Echo – Input pin **D6**
Trig - Output pin **D7**

```
int trigPin = 6;
int echoPin = 7;
long duration, cm;
int led = 3;

void setup( ) {
  pinMode(echoPin, INPUT);
   pinMode(trigPin, OUTPUT);
   pinMode(led, OUTPUT);
}
void loop( ) {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
```

Ultrasonic Sensor code

```
int trigPin = 6;
int echoPin = 7;
long duration, cm;
int led1 = 3;

void setup( ) {
  pinMode(echoPin, INPUT);
   pinMode(trigPin, OUTPUT);
   pinMode(led1, OUTPUT);
}
void loop( ) {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
```

```
duration = pulseIn(echoPin, HIGH);
cm = duration/29/2;

If(cm < 25){
      digitalWrite(led1, HIGH);
   }
   else {
      digitalWrite(led1, LOW);
  }
delay(1000);

}
```

Thank you!

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Japanese
IT University